

WEEK#09**ASCII & BCD INSTRUCTIONS****I. BCD Addition and Subtraction using Assembly Language**

The 8086/88 Assembly language provides correction of BCD addition and subtraction using instructions DAA and DAS.

a) DAA (Decimal Adjust for Addition)

This instruction corrects the result of BCD addition. It is used after ADD or ADC instruction with AL as destination register. It has no operand. The result of BCD addition must be stored in AL before application of DAA as it works only on AL register. For example;

```
MOV AL, 45H
MOV BL, 35H
ADD AL, BL
DAA
```

b) DAS (Decimal Adjust for Subtraction)

This instruction corrects the result of BCD subtraction. It is used after SUB or SBB instruction with AL as destination register. It has no operand. The result of BCD subtraction must be stored in AL before application of DAS as it works only on AL register. For example;

```
MOV AL, 45H
MOV BL, 35H
SUB AL, BL
DAS
```

II. ASCII Addition and Subtraction using Assembly Language

The 8086/88 Assembly language has provisions for adding and subtracting numbers without masking the higher 4 bits (i.e., 0011) using AAA and AAS.

a) AAA (ASCII Adjust for Addition)

It requires the result of addition to be held in AL register. AAA also works on unpacked BCD data in addition to ASCII. For example;

```
MOV AH, 00      ;AH=00
MOV AL, '7'      ;AL=37H
MOV BL, '5'      ;BL=35H
ADD AL, BL       ;AL=6CH
AAA              ;changes 6CH to 02 in AL and AH=CF=1
OR AX, 3030H     ;AX=3132 =>12H in ASCII
```

b) AAS (ASCII Adjust for Subtraction)

It requires the result of subtraction to be held in AL register. AAS also works on unpacked BCD data in addition to ASCII. For example;

```
MOV AX, 105H    ;AX=0105 =>unpacked BCD for 15
MOV CL, 06      ;CL=06H
SUB AL, CL      ;5 - 6 = -1 (FFH)
ADD AL, BL      ;AL=6CH
AAS             ;FFH in AL is adjusted to 09, and
                ;AH is decremented, leaving AX = 0009
```

III. ASCII Multiplication and Division using Assembly Language

The 8086/88 Assembly language has provisions for multiplication and division for numbers represented in ASCII using AAM and AAD. It is also applicable for unpacked BCD numbers.

a) AAM (ASCII Adjust for Multiplication)

It is meant for adjustment of unpacked BCD multiplication. Using AAM, the result of multiplication of two unpacked BCD numbers can be converted back to BCD. AAM works only on AX. For example;

```
MOV AL, '7'     ;AL = 37H
AND AL, 0FH     ;AL = 07 (unpacked BCD)
MOV DL, '6'     ;DL = 36H
AND DL, 0FH     ;DL = 06 (unpacked BCD)
MUL DL          ;AX = AL x DL = 07 x 06 = 002AH = 42
AAM             ;AX = 0402 (7x6=42 unpacked BCD)
OR AX, 3030H    ;AX = 3432 result in ASCII
```

b) AAD (ASCII Adjust for Division)

It is meant for adjustment of unpacked BCD division. Before dividing the unpacked BCD by another unpacked BCD, AAD is used to convert it to HEX; resulting in both the quotient and remainder in unpacked BCD. AAD works only on AX. For example;

```
MOV AX, 3539    ;AX = 3535 => ASCII for 59
AND AX, 0F0FH   ;AH = 05, AL = 09 unpacked BCD data
AAD            ;AX = 003BH hex equivalent of 59
MOV BH, 08H     ;divide by 08
DIV BH          ;3B / 08 gives AL = 07, AH = 03
OR AX, 3030H    ;AL = 37H (quotient) AH = 33H (remainder)
```